Cross-Domain Few-Shot Hyperspectral Image Classification With Cross-Modal Alignment and Supervised Contrastive Learning

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***Abstract*— Recently, metric-based few-shot learning (FSL) methods have achieved good performance in the hyperspectral image (HSI) classification. However, existing methods suffer from two problems: over-reliance on image modality information leads to inaccurate prototype representation, where a prototype refers to the centroid of each class in the dataset, and the impact of redundant and noisy pixels on model discriminability is rarely considered. These problems result in insufficient discriminability of the model for the target domain. To address the above issues, we propose a cross-domain few-shot HSI classification framework with cross-modal alignment and supervised contrastive learning (CDFS-CASCL). It is well known that human visual learning greatly benefits from the input of various modal information such as vision, language, and video. Inspired by the way humans abstract image class concepts in language form and understand the essence of classes, we perform cross-modal alignment (CA) between similar image and text prototypes and use abstract text semantics to guide the model to learn semantic-related features with good generalization ability in images, so as to improve the accuracy of image prototypes representation of the prototypes. In addition, through supervised contrastive learning (SCL) based on neighborhood pixel mask in the target domain, the enhanced sample features belonging to the same class are closer, while the enhanced sample features belonging to different classes are pulled further, enabling the model to learn mask-robust discriminative feature representations, suppressing the negative impact of redundant and noisy pixels, and improving the model's discriminability. The experimental results demonstrate the superiority of the proposed CDFS-CASCL. The code is available at https://github.com/Li-ZK/CDFS-CASCL-2024.**

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***Index Terms*— Cross-modal alignment (CA), few-shot learning (FSL), hyperspectral image (HSI) classification, supervised contrastive learning (SCL).**

I. Introduction

HYPERSPECTRAL images (HSIs) have rich information and subtle spectral differentiation capabilities. Therefore, they are widely used in mineral identification, military target detection, precision agriculture, and other fields. Achieving accurate classification of HSI is crucial for analyzing HSI. In the past few decades, researchers have continuously focused on HSI classification problems and have proposed many effective algorithms. Early classification algorithms such as k-nearest neighbor [1] and support vector machine [2] mainly considered the spectral information in HSI data. Later classification algorithms such as extended morphological contour [3] and Gabor wavelet [4] fully considered spatial–spectral features to generate better classification results [5].

Artificial intelligence algorithms and their increasingly important role in remote sensing data analysis [6]. In recent years, deep learning-based HSI classification methods have attracted the attention of researchers [7], [8]. Li et al. [9] developed an HSI classification framework using a 3-D convolutional neural network (3DCNN). Zhong et al. [10] proposed a spectral–spatial residual network (SSRN), which combines spectral and spatial residual blocks to learn highly discriminative features. Fu et al. [11] proposed a novel 3-D singular spectrum analysis (3DSSA) method to extract features to address intraclass spectral variability and spatial heterogeneity issues. Xu et al. [12] proposed a novel dual-channel residual network (DCRN) to resolve HSI classification with noisy labels. Feng et al. [13] used the GAN model for HSI classification and proposed a multicomplementary GAN (CMC-GAN) with contrastive learning to solve the modal collapse problem existing in the single-generation mode. He et al. [14] studied transformer for HSI classification and proposed a spatial spectrum transformer (SST) framework to solve the problem of capturing continuous spectral relationships in HSI over long distances. Chen et al. [15] proposed an HSI classification generation framework (SpectralDiff) with a diffusion model, which effectively captures the spectral–spatial distribution of data and the relationship between samples. Although the above

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methods have achieved fruitful results, training a deep learning model typically requires a large number of labeled samples. However, due to the high cost of collecting and labeling HSI, only limited labeled samples can be obtained. Therefore, how to use extremely limited labeled samples to train a deep model with good generalization ability has become an urgent problem to be solved.

To address the issue of scarce labels, researchers have proposed various strategies for small-sample learning. One method of data augmentation involves directly applying a series of transformations or adding random noise to expand the number of labeled samples [16], [17]. Semi-supervised deep learning methods solve HSI classification with limited labeled samples by fully exploiting unlabeled samples [18], [19]. Self-supervised learning methods extract supervised information from a large amount of unlabeled data to train the network and then fine-tune the network using a small amount of labeled information [20], [21]. As an important branch of self-supervised learning, contrastive learning methods can enable the model to obtain robust feature representation [22], [23], [24], [25]. Incremental learning is a method proposed for the emerging new classes in HSI classification, aiming to achieve the ability to quickly learn new classes through a small number of samples while maintaining good performance on the original classes [26], [27]. Recently, few-shot learning (FSL) has sparked great interest among scholars due to its ability to quickly generalize to new tasks with prior knowledge and limited labeled samples [28].

Most FSL methods utilize an episodic training strategy for meta-learning (i.e., learning to learn) [29], [30], [31], [32], aiming to learn a model from a set of seen tasks that can be generated to new tasks with few labeled training samples. In FSL, a large number of training episodes (known as meta-tasks) are constructed from a meta-training set containing abundant training samples. In each meta-task, the training data is divided into a support set and a query set, with the support set containing N classes and each class having K samples, and the query set having the same N classes. By training on various meta-tasks across different classes, the model can be generated during the meta-testing phase to classify unseen classes. Model-based, metric-based, and optimization-based approaches are employed to address few-shot classification challenges [33], with metric-based methods being commonly utilized in HSI classification for their simplicity and effectiveness. Li et al. [34] introduced a deep cross-domain FSL (DCFSL) approach for HSI classification that addresses the challenges of FSL and interdomain differences within a unified framework. Wang et al. [35] proposed a novel heterogeneous FSL (HFSL) method for HSI classification, which adopts learning transferable spatial structural and texture knowledge from natural images to improve classification accuracy. Zhang et al. [36] proposed to combine FSL with domain alignment based on graph information aggregation to reduce domain differences and better transfer base class knowledge. Li et al. [22] introduced self-supervised learning into the HSI field to solve the FSL classification problem, and proposed the FSCF-SSL method to better transfer meta-knowledge between natural images and HSIs. Liu et al. [37] imposed triple constraints to address the instability issue of support set prototypes, and adopted a fusion training strategy to mitigate the domain shift problem.

Although metric-based FSL methods have achieved good classification performance, the problem of weak model generalization performance due to insufficient labeled samples has not been well resolved. On the one hand, most of the current FSL methods used for HSI classification heavily rely on visual modal data, unable to capture semantic attributes from a very small number of samples, resulting in the lack of semantic representation capability in image prototypes. However, humans can extract valuable information from various modes of input such as visual and language [38]. Inspired by this, recent researchers have introduced multimodal information into few-shot classification. Xing et al. [39] considered that visual or semantic features may have different discriminative abilities for different concepts and proposed an adaptive mechanism that integrates information from two modalities based on newly learned image classes. Afham et al. [40] regularized visual features by forcing prototypes to retain semantic information about class descriptions, improving their generalization ability for novel classes in inference. Nishida et al. [41] solved the few-shot image classification task by learning new concepts from the linguistic description of the image, confirming that machine- and user-generated descriptions help improve classification accuracy. Afham and Rodrigo [42] introduced an alignment mechanism for visual and semantic feature vectors into existing FSL methods to learn more general visual concepts. In [43], FSL is incorporated with a transformer for improved HSI classification. Although these methods have achieved good results, they require rich class-level natural language descriptions for images, which are lacking in HSIs themselves. Instead, the class label text can be used directly. Therefore, we use the pretrained language model to extract features from the label text and obtain the semantic prior knowledge of the class to guide the learning of image prototype representation, so that the image feature extractor can learn a more representative prototype representation.

On the other hand, existing small-sample HSI classification methods rarely consider the impact of redundant and noisy pixels on class separability. The existing methods usually take a neighborhood cube containing the central pixel as a sample and use the class label of the central pixel as the class label of the sample, which will result in the generation of redundant and noisy pixels. Redundant pixels refer to pixels in a neighborhood cube that have the same class as the central pixel and highly consistent spectral information, while noisy pixels refer to pixels in a neighborhood cube that have different classes from the central pixel and do not have stable neighborhood relationships. Due to the limited labeled samples in the target domain, the model is more susceptible to overfitting caused by redundant or noisy pixels. To address this issue, it is necessary to suppress redundant and noisy pixels so that the model can extract more robust representations from neighborhood cubes.

Recently, the revival of contrastive learning has led to significant progress in self-supervised representation learning [44], [45], [46], [47], [48], [49], [50]. Self-supervised contrastive

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learning (SCL) can enable the model to obtain robust feature representations. Scholars have already introduced self-SCL into the field of small-sample HSI classification [21], [22], [23]. Zhao et al. [21] developed an integrated image augmentation module for multiband HSI, combining a self-SCL method to extract robust spatial–spectral features from a large number of unlabeled samples. Ma et al. [23] used SimSiam [48] contrastive learning structure in a novel way to improve the quality of feature representations after enhancing spectral and spatial data, respectively. Although the above two methods have learned higher quality feature representations, the data augmentation methods are still limited to traditional methods such as random rotation and flipping in the field of natural image classification, which cannot effectively solve the problems of redundant and noisy pixels in HSI data. Li et al. [22] proposed self-SCL using dropout augmentation on the features of limited labeled samples in the target domain. Although this feature-level augmented self-SCL method learns more discriminative individual knowledge, it cannot solve the problem of redundant and noisy pixels at the data level. In recent years, image masking, as a new data augmentation method, has been widely used in self-supervised representation learning [51], [52], [53]. Wei et al. [51] proposed a mask feature (MaskFeat) prediction method for self-supervised pretraining of video models. He et al. [52] proposed masked autoencoders (MAEs) to mask random patches of input images and reconstruct missing pixels as a self-supervised task. Xie et al. [53] systematically studied how to learn good representations for masked image modeling tasks and proposed a simple framework for masked image modeling (SimMIM). Inspired by the preceding work, the self-SCL of image masks can alleviate the negative impact of redundant and noisy pixels on feature extraction to a certain extent. However, due to the lack of labels, positive examples in self-SCL typically consist of data augmentation samples that are homologous to the anchor, while negative examples consist of augmentation samples from other samples in a mini-batch. If used directly for classification tasks, potential similar samples in negative examples will be pulled away, which is not conducive to improving class separability. Khosla et al. [54] introduced label information into self-SCL, and achieved good results. Unlike self-SCL which only uses a single positive sample, this method not only considers multiple negative samples but also selects samples with the same anchor class as positive samples. Introducing supervisory information can make the embedded features of the same class closer than those of different classes, thereby enhancing class separability. Therefore, combined with the characteristics of HSI data, we adopt SCL based on neighborhood pixel masks. This is not only conducive to learning a mask-robust feature representation but also takes into account the separability of classes, which can generate mask-robust discriminative feature representation.

In summary, in order to enhance the semantic representation ability of image prototypes and suppress the impact of redundant and noisy pixels on model discriminability, this article proposes a cross-domain few-shot HSI classification method (CDFS-CASCL) with cross-modal alignment (CA) and SCL. First, we regard label text information as text modality and use it to guide the learning of image prototype representation by aligning text prototypes and image prototypes, thereby improving the semantic representation ability of image prototypes. Then, combining the characteristics of HSIs, we designed a SCL strategy based on neighborhood pixel mask to learn mask-robust discriminative feature representations, in order to suppress the influence of redundant and noisy pixels and improve the separability between classes.

The main contributions of this article can be summarized as follows.

1) We propose a cross-domain few-shot HSI classification framework with CA and SCL (CDFS-CASCL), which can learn a classification model suitable for limited labeled samples.
2) A CA method is designed to make the label text prototypes and image prototypes closer in the multimodal embedding space, which helps the image feature extractor learn the semantic prior knowledge of the labels.
3) A SCL strategy based on neighborhood pixel mask is proposed to minimize the distance of all positive pairs relative to negative pairs to learn a mask-robust discriminative feature representation. This suppresses the influence of redundant and noisy pixels and improves classification accuracy.

The remaining sections of this article are organized as follows. Section II details the proposed CDFS-CASCL method. Section III presents extensive experimental results and analyses. Finally, the conclusion is drawn in Section IV.

II. Proposed Method

Let a source domain dataset be denoted as D_s which contains C_s classes and a target domain dataset be D_t with C_t classes. In a regular setting [34], to ensure the diversity of training samples, C_s is usually larger than C_t. In addition, the classes in D_s are called base classes, and the classes in D_t are called novel classes. All of D_s are labeled samples, while D_t contains a small number of labeled samples D_l and a large number of unlabeled samples D_u. Among them, the samples of D_s and D_l are used for the training of the base class and the novel class, respectively, and the sample of D_u is used for the testing of the novel class. Since there are too few labeled samples in D_l, it is not enough to construct the meta-task well. Therefore, we use random Gaussian noise for data augmentation to expand the number of labeled samples of each class. The proposed CDFS-CASCL method uses the source domain dataset D_s and the target domain limited labeled dataset D_l to train the image feature extractor, and finally in the target domain dataset D_u to test its classification performance.

A. CDFS-CASCL Framework

Fig. 1 depicts the flowchart of the CDFS-CASCL network, including image feature extraction module, text feature extraction module, source domain and target domain FSL module, CA module, and mask-based SCL module. The training phase mainly consists of three parts: first, FSL is performed on the source domain and target domain. The general discriminative knowledge of the source domain can be transferred to the

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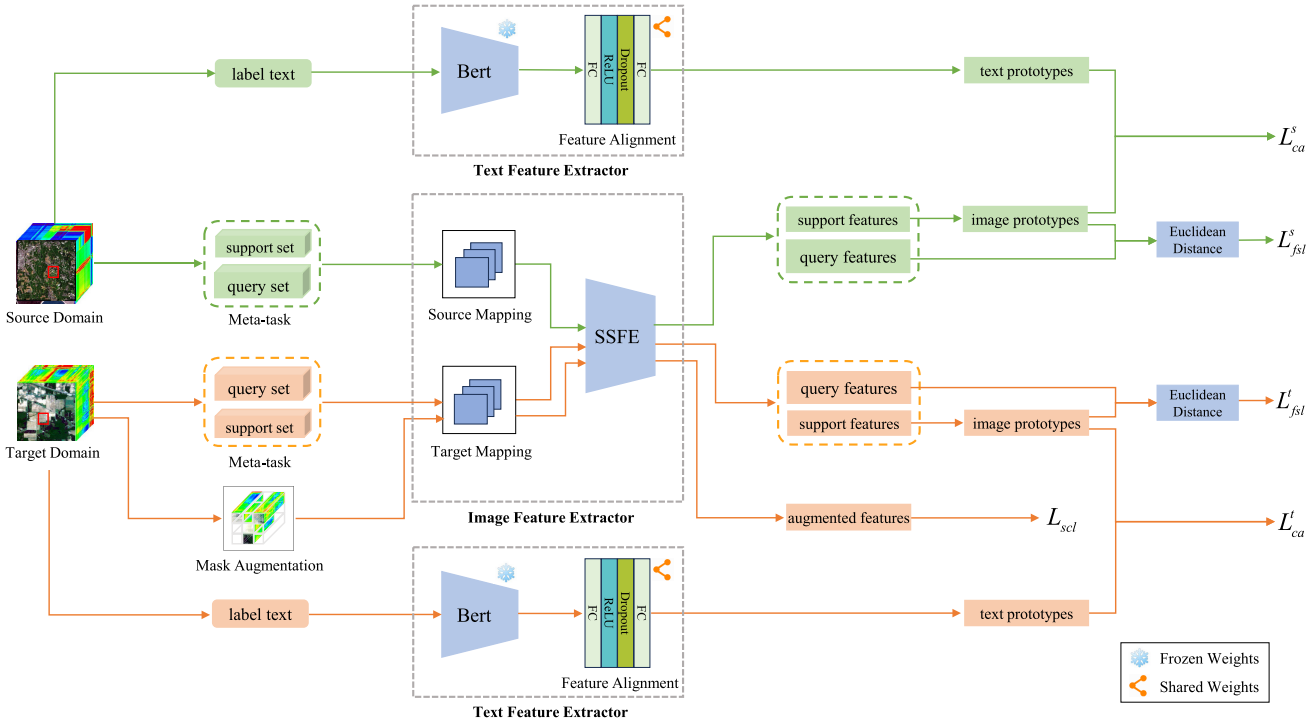


Fig. 1. Framework of the proposed CDFS-CASCL method. It mainly includes five modules: image feature extraction module, text feature extraction module, source domain and target domain FSL module, CA module, and mask-based SCL module. The training phase mainly consists of three parts: first, construct a meta-task for FSL to transfer common discriminative knowledge in the source domain to the target domain. Second, the CA module aligns text prototypes and image prototypes to guide the image feature extractor to learn semantic prior knowledge. Finally, the mask-based SCL module performs SCL based on neighborhood pixel mask on the target domain samples to learn a mask-robust discriminative feature representation and suppress the impact of redundant and noisy pixels on class separability.

target domain through the shared image feature extractor. Next, we align the image and label text prototypes of each class in the source and target domains. Label text prototypes and image prototypes are closer in the multimodal embedding space, which helps the image feature extractor obtain semantic prior knowledge about the label text. Finally, we perform SCL on the target domain samples. By reducing the distance between positive samples of the same class and increasing the distance between negative samples of different classes, it can learn mask-robust discriminative feature representation, suppress the influence of redundant and noisy pixels on class separability, and improve the ability to distinguish novel class samples in the target domain. Combining the above three parts, the total loss function of the model in the training stage can be expressed as

$$L = L_{\text{fsl}} + \lambda_1 L_{\text{ca}} + \lambda_2 L_{\text{scl}} \tag{1}$$

where L_{fsl} represents the total FSL classification loss of the model, L_{ca} represents the total CA loss of image modality and text modality, and L_{scl} represents SCL loss of the target domain. Here, λ_1 and λ_2 are regularization parameters. The overall training phase is shown in Algorithm 1.

During the testing phase, we only use the trained target mapping layer and image feature extractor to extract image embedding features for all testing samples. We then train the nearest neighbor classifier using a small number of labeled samples of the novel class. In the embedding space, since the image feature extractor can map sample features of the same class closer and map sample features of different classes farther, we can predict the class of the unlabeled testing samples by comparing its distance with the features of other labeled samples.

B. Image Feature Extractor

The image feature extractor includes a mapping layer and spatial–spectral feature extractor (SSFE), which is used to encode features of the original image to obtain image features. First, the mapping layers, including the source domain mapping layer and target domain mapping layer, use the mapping layer first proposed in [55] to ensure that input samples with different channel numbers in the source domain and target domain have the same dimensions to share image feature extractor. The mapping layer is essentially a 2-D convolutional layer consisting of d convolution kernels of size $1 \times 1 \times c$, where c and d are the dimensions of the input and output data, respectively. For source domain and target domain datasets with different input dimensions c, the dimension of the feature map channel output through the corresponding mapping layer is the same, that is d.

The SSFE is shown in Fig. 2, which includes a spatial feature extraction module, a spectral feature extraction module, and a spatial–spectral feature fusion module. It is used to extract discriminative spatial–spectral embedding features from input samples. Similar to [12], we perform spatial feature extraction and spectral feature extraction on the input samples separately. Differently, we map both spatial and spectral

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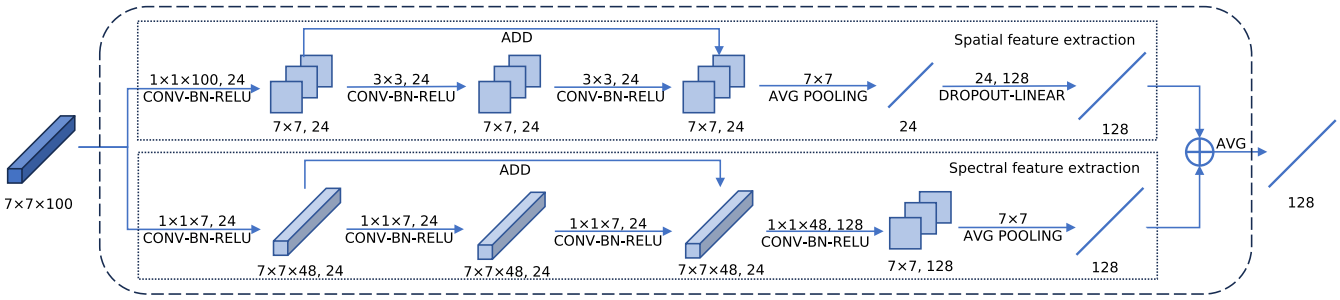


Fig. 2. Architecture diagram of SSFE. SSFE includes a spatial feature extraction module, a spectral feature extraction module, and a spatial–spectral feature fusion module, which is used to extract the discriminative spatial–spectral embedding features of the input sample. First, spatial feature extraction and spectral feature extraction are performed on the input samples simultaneously. Second, both spatial and spectral features are mapped into an embedding space of the same dimension. Finally, the spatial features and spectral features are fused with equal weights to obtain the spatial–spectral joint features.

Algorithm 1 Training Process of CDFS-CASCL

Input:

The image feature extractor f_φ
The text feature extractor f_θ
The support set S_s and query set Q_s in D_s
The support set S_t and query set Q_t in D_l
The source domain label text set T_s
The target domain label text set T_t
Number of training episodes E

Output:

The total loss L of the model

for episode in $\{1, \ldots, E\}$ **do**
  Train with S_s, Q_s, S_t, Q_t, T_s and T_t
  Calculate image prototype of $S_s \rightarrow C_s$, $S_t \rightarrow C_t$
  Calculate text prototype of $T_s \rightarrow S_s$, $T_t \rightarrow S_t$
  for $(x_j^s, y_j^s) \in Q_s$ **do**
    Calculate the L_{fsl}^s of an episode by Ep. (4)
  end for
  Calculate the L_{fsl}^t of an episode same as L_{fsl}^s
  for $x_i^s \in S_s, t_i^s \in T_s$ **do**
    Calculate the L_{ca}^s of an episode by Ep. (7)
  end for
  Calculate the L_{ca}^t of an episode same as L_{ca}^s
  for $(x_i^t, y_i^t) \in D_l$ **do**
    Calculate the L_{scl} of an episode by Ep. (11)
  end for
  Return the total Loss L of an episode by Ep. (1)
end for

features to the same dimensional embedding space and then fuse spatial and spectral features with equal weights to obtain spatial–spectral joint features.

1) Spatial Feature Extraction Module: The spatial feature extraction module is used to extract spatial neighborhood information of target pixels. The spatial feature extraction module is constructed by a convolutional layer, a residual block, and a fully connected layer. The residual block contains two convolutional layers, each of which is followed by a batch normalization (BN) layer and a ReLU activation function. The convolution layer uses 24 convolution kernels with a size of $1 \times 1 \times 100$, and its sampling step is (1, 1, 1), which is used to compress multiple channel spectral features into a single channel while retaining spatial information. The goal is to make the network focus more on the spatial domain. The residual block consists of two identical convolutional layers containing 24 convolution kernels of size (1, 3, 3), which are used to enhance attention to important spatial regions while preventing network degradation. The residual block is followed by a 7×7 global average pooling to integrate spatial information to obtain a feature vector representation. Dropout is added before the fully connected layer to prevent the model from overfitting. The fully connected layer structure is (24, 128), which maps the feature vector to 128 dimensions consistent with the spectral dimensions.

2) Spectral Feature Extraction Module: The spectral feature extraction module is used to extract spectral information of target pixels. The spectral feature extraction module contains two convolutional layers and a residual block. The residual block contains two convolutional layers, each of which is followed by a BN layer and a ReLU activation function. The first convolutional layer uses 24 convolution kernels with a size of $1 \times 1 \times 7$ and a sampling step of (1, 1, 2) to eliminate redundant spectral features. The residual block consists of two consecutive identical convolutional layers, each containing 24 convolutional kernels of size $1 \times 1 \times 7$ with a sampling step size of (1, 1, 1), to enhance the model's attention to important channels while preventing network degradation. The second convolution layer uses 128 convolution kernels with a size of $1 \times 1 \times 48$, and its sampling step is (1, 1, 1) to achieve feature conversion and dimensionality reduction and to achieve alignment with the spatial feature dimensions. Finally, a 128-D spectral feature vector is obtained through 7×7 global average pooling.

3) Spatial–Spectral Fusion Module: The spatial–spectral fusion module averages the spatial and spectral types of features as a spatial–spectral joint feature. The purpose of equal weight fusion is to balance the importance of the two types of information and integrate the discriminability of spatial features and spectral features, thereby further expanding the differences between classes and enhancing the classification performance of the network.

C. Text Feature Extractor

The text feature extractor includes the Base Bert model and the feature alignment layer, which is used to perform feature encoding on the input label text to obtain text features.

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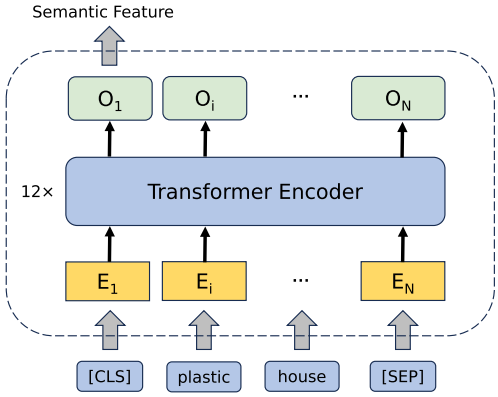


Fig. 3. Architecture diagram of Base Bert. [CLS] is a special classification token. [SEP] is a special separator token. E_i represents the embedding at position i of the input sequence. O_i is the last hidden state at position i of the input sequence. Among them, the final hidden state O_1 corresponding to the [CLS] token is the semantic feature of the input label text. For example, we split the label text "plastic house" into two tokens: "plastic" and "house," and send them to Base Bert as a sentence, and obtain the semantic feature of the label text through the O_1.

1) Base Bert Model: The basic structure of the Base Bert [56] model is the encoder part of a standard transformer [57] structure. A standard transformer encoder is stacked by multihead-attention, layer normalization, and feedforward. Each layer of Bert consists of one of such transformer encoder units. The Base Bert model used in this article is stacked by 12 such hierarchical structures. Each layer has 12 attentions. The word vector dimension H is 768. The size of the feedforward is set to $4H$, which is 3072. Therefore, we use Base Bert to generate the output dimension of each class label text to be 768.

The Base Bert framework used in this article is shown in Fig. 3. The input label text is treated as a sentence and decomposed into a sequence of tokens. In addition, the sequence contains two special markers [CLS] and [SEP]. The first token [CLS] is a special classification token, and [SEP] is a special separator token used to separate different sentences. Let E_i represent the embedding at position i of the input sequence, which is the sum of token embedding, segmentation embedding, and position embedding, and O_i be the last hidden state at position i of the input sequence. Among them, the final hidden state O_1 corresponding to the [CLS] token is used as the aggregated sequence representation for the classification task. When we take the label text as input, O_1 is the semantic feature representation of the input label text.

2) Feature Alignment Layer: The feature alignment layer is used to map the features output by the Base Bert model to the same dimensions as the image features to facilitate subsequent cross-modal feature alignment operations. The feature alignment layer is stacked by linear, ReLU, dropout, and linear layers. The first linear maps the 768-D original text feature to the same 128-D feature as the image feature. The second linear is a dimensionally invariant feature map, which is used to further extract and combine higher level feature representations to enhance the expression ability of the model.

D. Source and Target FSL

In order to better learn the general discriminative knowledge in the source domain and ensure that the features learned by the image feature extractor have good separability in the embedding space, we conducted episode training in meta-learning on the shared image feature extractor in the source and target domains, which we refer to as source domain FSL and target domain FSL.

Taking the source FSL as an example, each episode represents a meta-task, consisting of a support set and a query set. During meta-training, we randomly select N classes from the source dataset, where each class has n_k labeled samples as the support set: $S_s = \{(x_i^s, y_i^s)\}_{i=1}^{N\times n_k}$, and n_q unlabeled samples as the query set: $Q_s = \{(x_j^s, y_j^s)\}_{j=1}^{N\times n_q}$. The image prototype of the cth class in the support set is calculated as follows:

$$c_k^s = \frac{1}{N}\sum_{x_i^s \in C_k^s} f_\varphi\left(x_i^s\right) \tag{2}$$

where C_k^s represents the set of support set samples in the kth class of the source domain, and c_k^s represents the image prototype of the kth class sample in the source domain obtained by averaging the features of the kth class support set samples.

The probability of a query sample x_j in the query set Q_s belonging to the kth class is computed as follows:

$$P\left(y_j^s = k | x_j^s \in Q_s\right) = \frac{\exp\left(-D\left(f_\varphi\left(x_j^s\right),\ c_k^s\right)\right)}{\sum_{k=1}^{N}\exp\left(-D\left(f_\varphi\left(x_j^s\right),\ c_k^s\right)\right)} \tag{3}$$

where $D(,)$ represents a distance metric function for measuring the distance between two vectors, which is the Euclidean distance function in this research, $f_\varphi(\cdot)$ represents the source mapping layer and the shared image feature extractor, c_k represents the image prototype of the kth class, N represents the number of classes in each episode, and x_j and y_j represent the feature and corresponding label of the query sample, respectively. The FSL loss for each episode of the source domain can be computed using negative log probability as follows:

$$L_{\text{fsl}}^s = E_{S_s,Q_s}\left[-\sum_{(x,y)\in Q_s}\log p(y=k|x)\right]. \tag{4}$$

The FSL loss for each episode of the target domain L_{fsl}^t is similar to the source domain. Therefore, the overall FSL loss is given by

$$L_{\text{fsl}} = L_{\text{fsl}}^s + L_{\text{fsl}}^t. \tag{5}$$

E. Cross-Modal Alignment

Inspired by the widespread application of multimodal and contrastive learning methods, we adopt contrastive learning between image and text features to achieve alignment of image and text modalities to enhance FSL. Because the alignment of the two modalities will enable the image prototype to produce visual understanding with stronger generalization capabilities across modalities.

For each episode, the sets of source domain and target domain label text are T_s and T_t, respectively. Taking the source domain as an example, for each class $i \in N$, we have the corresponding class label text $t_i^s \in T_s$. We use a text

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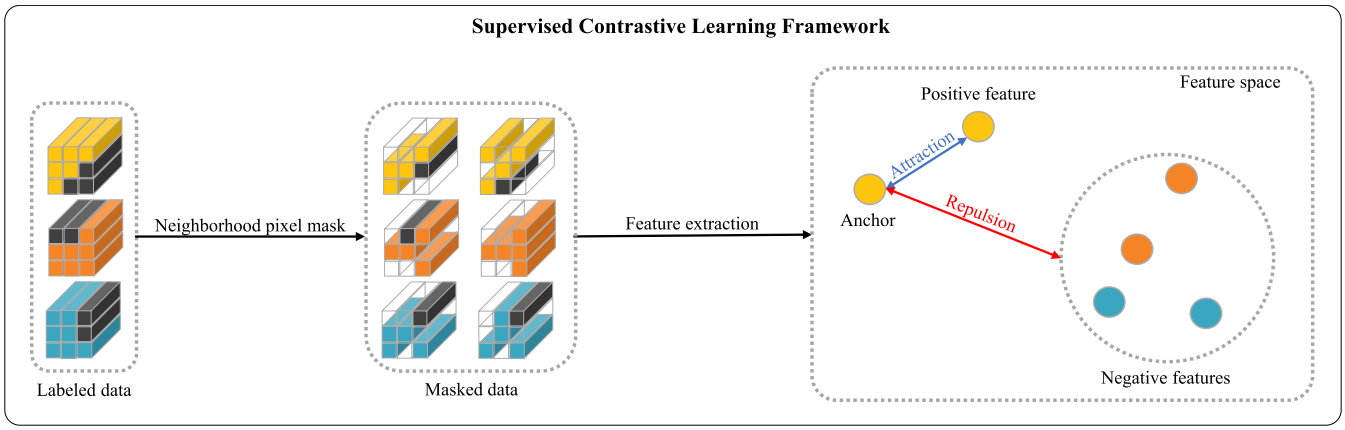


Fig. 4. SCL framework based on neighborhood pixel masks. Different colors are used in the figure to represent different classes, and black represents pixels of different classes from the center. First, neighborhood pixel masking is performed on a batch of labeled data. Second, by extracting the features of the masked data. Finally, each feature is used as an anchor in turn, and the SCL loss is calculated by bringing the MaskFeats of the same class of anchor closer and moving the MaskFeats of different classes farther away.

feature extractor f_θ, including Base Bert and feature alignment layers. The Base Bert uses the pretrained model parameters Base Bert uncased for parameter initialization and then freezes the network weights to extract the original text features. The feature alignment layer is used to map the original text features to the same dimensions as the image features to obtain the final text features s_i^s, $s_i^s = f_\theta(t_i^s)$. Treat the text features s_i^s in each task as the text prototypes of the class. Then, we use the NT-Xent loss function introduced in [46] to modally align the image and text prototypes of each class.

For an N-way one-shot meta-task, the image and text prototype sets are, respectively, expressed as $C_s = \{c_1^s, c_2^s, \ldots, c_N^s\}$ and $S_s = \{s_1^s, s_2^s, \ldots, s_N^s\}$, so c_i^s and s_i^s are two modes of the same sample, which is positive sample of each other, and other samples are negative samples. The objective of contrastive learning is to learn a feature extractor that increases the similarity between positive sample pairs and reduces the similarity between negative sample pairs. The source domain alignment loss L_{cl}^s based on NT-Xent loss can be expressed as

$$L_{\text{cl}}^s\left(c_i^s, s_i^s\right) = -\log \frac{\exp\left(\text{sim}\left(c_i^s, s_i^s\right)/\tau\right)}{\sum_{k=1,k\neq i}^{N}\exp\left(\text{sim}\left(c_i^s, c_k^s\right)/\tau\right)+\sum_{k=1}^{N}\exp\left(\text{sim}\left(c_i^s, s_k^s\right)/\tau\right)} \tag{6}$$

where sim(,) is a similarity measurement function between c_i^s and s_i^s, which is the negative Euclidean distance here, and τ is the temperature coefficient that controls the discrimination of negative samples. Considering that $L_{\text{cl}}^s(c_i^s, s_i^s) \neq L_{\text{cl}}^s(s_i^s, c_i^s)$, the average contrastive loss within an episode in the source domain can be expressed as

$$L_{\text{ca}}^s = \frac{1}{2N}\sum_{i=1}^{N}\left(L_{\text{cl}}^s\left(c_i^s, s_i^s\right) + L_{\text{cl}}^s\left(s_i^s, c_i^s\right)\right). \tag{7}$$

Therefore, the overall average contrast loss in an episode can be expressed as

$$L_{\text{ca}} = L_{\text{ca}}^s + L_{\text{ca}}^t. \tag{8}$$

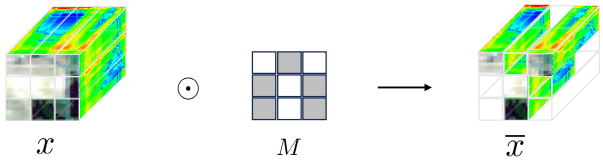


Fig. 5. Masking process of sample x by the proposed neighborhood pixel mask.

F. Mask-Based SCL

SCL works by maximizing the consistency between augmented views of samples of the same class and the difference between augmented views of different classes. It aims to reduce the distance between samples of the same class in the embedding space while increasing the distance between samples of different classes. The data augmentation method of the model is crucial to the learning of model augmentation invariance. Therefore, this study chooses to perform SCL based on neighborhood pixel mask on the target domain, as shown in the framework in Fig. 4. This framework iteratively brings masked samples of the same class closer while pulling samples of different classes farther away, allowing the model to focus on all discriminative related features in the unmasked area, not just the most discriminative features, as these features may be redundant or noisy pixels. Therefore, this method can enable the model to learn robust neighborhood discriminative features, reduce the model's excessive focus on redundant and noisy pixels, and thus prevent overfitting of the model. The proposed SCL based on neighborhood pixel mask consists of three parts: data augmentation, encoder network, and loss function.

1) Data Augmentation: For each labeled sample x in the target domain, data augmentation is performed twice to obtain two relevant views of the same sample. In this article, we propose to use neighborhood pixel masks for data augmentation and to learn mask-robust discriminative features to suppress the influence of redundant and noisy pixels. Samples in HSI are represented as patches, with dimensions of $W \times H \times C$. We randomly mask all pixels in the patch according to a fixed mask ratio σ to obtain mask-augmented samples. The neighborhood pixel mask process is shown in Fig. 5. Suppose

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x is an original sample, and the sample augmented by random pixel masks is expressed as $\overline{x}$, $\overline{x} = \mathrm{Aug}(x)$. The calculation process is as follows:

$$\overline{x} = x \odot M_{i,j} \tag{9}$$

where $\odot$ represents Hadamard product, and M is the mask template with size $W \times H$. Each element $M_{i,j}$ in the mask template M is calculated as

$$M_{i,j} = \begin{cases} 0, & \text{if } r \le \sigma \\ 1, & \text{if } r > \sigma \end{cases} \tag{10}$$

where r is a value randomly sampled from the uniform distribution of $[0, 1]$, and σ is the pixel mask ratio in the neighborhood pixel mask method.

2) Encoder Network: The encoder network uses the target domain image feature extractor f_φ^t, which includes the target domain mapping layer and SSFE. The augmented data $\overline{x}$ is fed into f_φ^t, and the corresponding embedded feature representation z is obtained, $z = f_\varphi(\overline{x}) \in R^d$, where d denotes the embedded feature dimension.

3) Loss Function: For all labeled samples in the target domain, we randomly select N data-label pairs $\{(x_i, y_i)\}_{i=1}^N$ to do SCL. By performing neighborhood pixel mask on the N samples, a set of $2N$ augmented samples $\{(\overline{x}_a, \overline{y}_a)\}_{a=1}^{2N}$ is generated, where $\overline{x}_{2i-1}$ and $\overline{x}_{2i}$ are two augmented views of x_i, $\overline{y}_{2i-1} = \overline{y}_{2i} = y_i$. The target domain SCL loss can be expressed as

$$L_{\mathrm{scl}} = \sum_{a \in I} \frac{-1}{|P(a)|} \sum_{p \in P(a)} \log \frac{\exp(z_a \cdot z_p/\tau)}{\sum_{l \in A(a)} \exp(z_a \cdot z_l/\tau)} \tag{11}$$

where z represents the embedding vector obtained by feature extraction of the sample, the $\cdot$ notation represents vector inner product, $\tau \in R^+$ is the temperature coefficient, controlling the discrimination of negative samples, $a \in I \equiv \{1, \ldots, 2N\}$ serves as the index of an enhanced sample, called an anchor. The set of indices in I other than a is $A(a) \equiv I \backslash \{a\}$. The positive sample index set corresponding to anchor a is $P(a) \equiv \{p \in A(a) : \overline{y}_p = \overline{y}_a\}$, and $|P(a)|$ is its cardinality. The elements in $A(a)$ except p are used as negative sample indexes.

III. Experiments

A. Experimental Datasets

We use five publicly available HSI datasets, including Chikusei (CH), Indian Pines (IP), Houston (HT), Salinas (SA), and LongKou (LK) [58] to evaluate the effectiveness of the proposed method. Following the FSL experimental settings in [34], CH is used as the source domain dataset, and the other four are used as the target domain datasets.

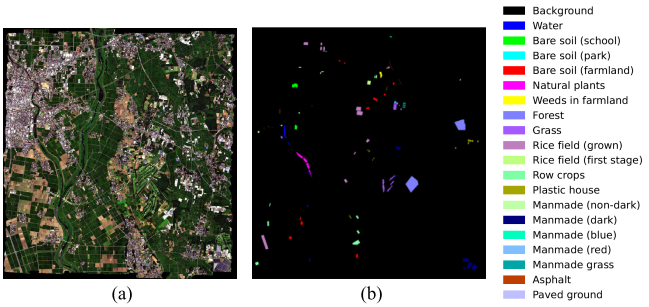
1) Source Domain Dataset: Chikusei: This dataset was captured by CH's Headwall Hyperspec-VNIR-C sensor and contains 2517 × 2335 pixels with a spatial resolution of 2.5 m/pixel. The dataset contains 19 land cover classes and 128 bands with a wavelength range of 343–1018 nm. Table I shows the number of samples in each class in the CH dataset. The pseudo-color image and corresponding ground-truth map of the CH dataset are shown in Fig. 6.

TABLE I
NUMBER OF SAMPLES FOR EACH CLASS IN THE CH DATASET

Class	Name	Number of Samples
1	Water	2345
2	Bare soil (school)	2859
3	Bare soil (park)	236
4	Bare soil (farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field (groim)	13369
10	Rice field (first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade (nan-dark)	1220
14	Manmade (dark)	7664
15	Manmade (blue)	431
16	Manmade (red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
Total		77592

TABLE II
NUMBER OF SAMPLES FOR EACH CLASS IN THE IP DATASET

Class	Name	Number of Samples
1	Alfalfa	46
2	Corn notill	1428
3	Corn mintill	830
4	Corn	237
5	Grass pasture	483
6	Grass trees in farmland	730
7	Grass pasture mowed	28
8	Hay windrowed	478
9	Oats	20
10	Soybean notill	972
11	Soybean mintill	2455
12	Soybean clean	593
13	Wheat	205
14	Woods	1265
15	Buildings Grass Trees Drives	386
16	Stone Steel Towers	93
Total		10249



(a) (b)

Fig. 6. Pseudo-color image and ground-truth map of CH. (a) Pseudo-color image and (b) ground-truth map.

2) Target Domain Dataset:

a) Indian Pines: IP dataset was collected by AVIRIS sensors at the IP test site in northwest Indiana and contains 145 × 145 pixels with a spatial resolution of 20 m/pixel. The dataset contains 16 land cover classes and 200 bands with wavelengths ranging from 400 to 2500 nm. Table II shows

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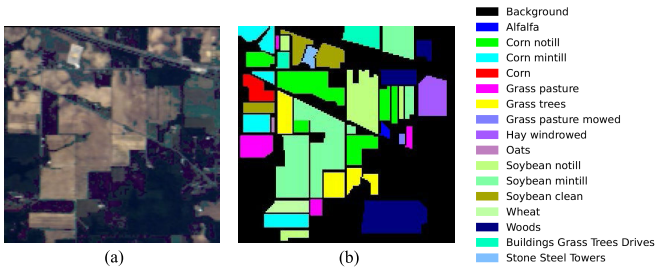


Fig. 7. Pseudo-color image and ground-truth map of IP. (a) Pseudo-color image and (b) ground-truth map.

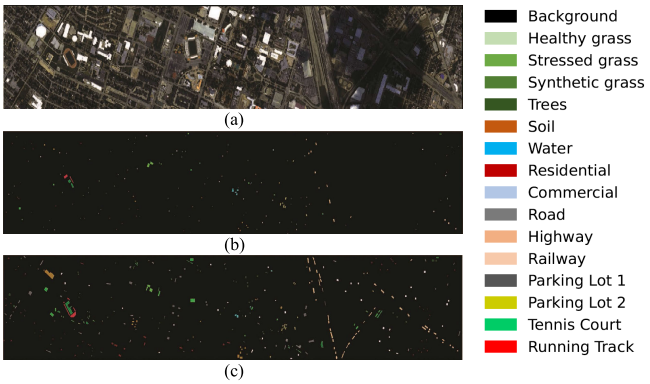


Fig. 8. Pseudo-color image and ground-truth map of HT. (a) Pseudo-color image, (b) ground-truth map of the training set, and (c) ground-truth map of the testing set.

the number of samples in each class in the IP dataset. The pseudo-color image and corresponding Ground-truth map of the IP dataset are shown in Fig. 7.

b) Houston: The HT dataset was introduced for the 2013 GRSS data fusion competition. The data was collected by airborne sensors over the University of HT campus and adjacent areas in 2012. It contains 349 × 1905 pixels with a spatial resolution of 2.5 m/pixel. The dataset contains 15 land cover classes and 144 bands, ranging from 364 to 1046 nm. Table III shows the number of training and testing set samples for each class in the HT dataset. The pseudo-color image of the HT dataset and the ground-truth maps of the corresponding training and testing sets are shown in Fig. 8.

c) Salinas: The SA dataset was collected by the AVIRIS sensor at the SA Valley Test Site in California and contains 512 × 217 pixels with a spatial resolution of 3.7 m/pixel. The dataset contains 16 land cover classes and 204 bands with wavelengths ranging from 400 to 2500 nm. Table IV shows the number of samples in each class in the SA dataset. The pseudo-color image and corresponding ground-truth map of the SA dataset are shown in Fig. 9.

d) WHU-Hi-LongKou: The LK dataset was acquired by a Headwall Nano-Hyperspec sensor installed on a UAV platform in the agricultural area of Longkou Town (Jingzhou City), Hubei Province, China. It contains 550 × 400 pixels with a spatial resolution of 0.463 m/pixel. The dataset contains nine land cover classes and 270 bands with wavelengths ranging from 400 to 1000 nm. Table V shows the number of samples in each class in the LK dataset. The pseudo-color image and corresponding ground-truth map of the LK dataset are shown in Fig. 10.

TABLE III
NUMBER OF TRAINING AND TESTING SET SAMPLES FOR EACH CLASS IN THE HT DATASET

Class	Name	Number of Training Set Samples	Number of Testing Set Samples
1	Healthy grass	198	1053
2	Stressed grass	190	1064
3	Synthetic grass	192	505
4	Trees	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot 1	192	1041
13	Parking Lot 2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
Total		2832	12197



Fig. 9. Pseudo-color image and ground-truth map of SA. (a) Pseudo-color image and (b) ground-truth map.

TABLE IV
NUMBER OF SAMPLES FOR EACH CLASS IN THE SA DATASET

Class	Name	Number of Samples
1	Brocoli green weeds 1	2009
2	Brocoli green weeds 2	3726
3	Fallow	1976
4	Fallow rough plow	1394
5	Fallow smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes untrained	11271
9	Soil vinyard develop	6203
10	Corn senesced green weeds	3278
11	Lettuce romaine 4wk	1068
12	Lettuce romaine 5wk	1927
13	Lettuce romaine 6wk	916
14	Lettuce romaine 7wk	1070
15	Vinyard untrained	7268
16	Vinyard vertical trellis	1807
Total		54129

B. Experimental Setups

The hardware configuration of the workstation used in the experiment is as follows: w7-3445 CPU (2.9 GHz), 256 GB memory, NVIDIA RTX6000 Ada GPU (48 GB).

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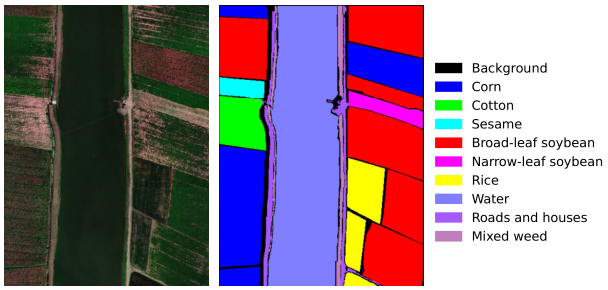


Fig. 10. Pseudo-color image and ground-truth map of LK. (a) Pseudo-color image and (b) ground-truth map.

TABLE V
NUMBER OF SAMPLES FOR EACH CLASS IN THE LK DATASET

Class	Name	Number of Samples
1	Corn	34511
2	Cotton	8374
3	Sesame	3031
4	Broad-leaf soybean	63212
5	Narrow-leaf soybean	4151
6	Rice	11854
7	Water	67056
8	Roads and houses	7124
9	Mixed weed	5229
Total		204542

The experimental code is implemented using the open-source software framework PyTorch.

In the CDFS-CASCL method, we select patches of size $7 \times 7 \times B$ as the input of the network model, where B is the number of bands. For the FSL of base and novel classes, the number of training episode iterations is set to 5000. Meta-task settings follow [34]. For each meta-task, n_k is set to 1, which means that each class takes 1 labeled sample as the support set. n_q is set to 19, which means that 19 unlabeled samples of each class are taken as the query set. The batchsize in SCL is set to 64. The temperature coefficient τ in L_{ca} and L_{scl} is both set to 0.1. Furthermore, our model is optimized using SGD with the learning rate set to 0.001 and the weight decay set to 0.0001. Overall accuracy (OA), average accuracy (AA), and kappa coefficient (KC) are used to evaluate the classification performance of all methods.

C. Parameter Tuning

In order to achieve better performance on the IP, HT, SA, and LK datasets, we performed parameter tuning on the dropout ratio η of the feature alignment layer, the mask enhancement ratio σ in the SCL module, the L_{ca} loss weight λ_1 and the L_{scl} loss weight λ_2. The dropout ratio η is selected from {0.1, 0.3, 0.5, 0.7, 0.9}, the mask enhancement ratio σ is selected from {0.2, 0.4, 0.6, 0.8}, the L_{ca} loss weight λ_1 is selected from {0.5, 1, 1.5, 2, 2.5}, and the L_{scl} loss weight λ_2 is selected from {0.5, 1, 1.5, 2, 2.5}. To alleviate the pressure of parameter optimization on different datasets, we adopted a suboptimal parameter adjustment strategy. First, other parameters adopt default values, then adjust the value of the first parameter. After determining the value of the first parameter, fix the parameter value, further adjust and determine the value of the second parameter, and so on. In addition, to mitigate the impact of randomness, the average of ten experimental results was reported for all different parameter combinations. Tables VI–IX use OA as the evaluation metric to display the experimental results of these four datasets for the parameters η, σ, λ_1, and λ_2, respectively. Based on the above experimental results, on the IP, HT, SA, and LK datasets, we set the parameters η to 0.3, 0.7, 0.5, and 0.1, and σ to 0.8, 0.2, 0.4, and 0.4, respectively. The parameter λ_1 is set to 2, and the parameter λ_2 is set to 2, 1, 1, and 1, respectively.

For the values of mask ratio σ in different datasets, combined with the ground-truth of the corresponding datasets mentioned above, it can be found that the IP dataset with rich neighborhood classes has the best effect when the mask ratio σ is high, while HT dataset with sparse neighborhood classes have the best effect when the mask ratio σ is low. SA and LK

TABLE VI
PARAMETER TUNING OF THE DROPOUT RATE η USING THE TARGET DOMAIN DATASETS

Dataset	Dropout ratio η				
	0.1	0.3	0.5	0.7	0.9
IP	81.45	**81.58**	81.41	81.41	77.49
HT	76.05	76.33	76.08	**76.63**	75.80
SA	94.05	94.25	**94.34**	94.22	93.68
LK	**96.27**	96.22	96.10	96.22	96.09

TABLE VII
PARAMETER TUNING OF THE MASK RATIO σ USING THE TARGET DOMAIN DATASETS

Dataset	Mask ratio σ			
	0.2	0.4	0.6	0.8
IP	81.02	81.05	81.29	**81.58**
HT	**76.63**	76.49	76.42	76.24
SA	94.25	**94.34**	94.26	94.26
LK	96.23	**96.27**	96.13	95.93

TABLE VIII
PARAMETER TUNING OF THE L_{ca} WEIGHT λ_1 USING THE TARGET DOMAIN DATASETS

Dataset	L_{ca} Weight λ_1				
	0.5	1	1.5	2	2.5
IP	81.16	81.46	81.55	**81.58**	81.42
HT	76.17	76.16	76.09	**76.63**	76.52
SA	94.10	94.10	94.20	**94.34**	94.32
LK	96.14	96.20	96.06	**96.27**	96.17

TABLE IX
PARAMETER TUNING OF THE L_{scl} WEIGHT λ_2 USING THE TARGET DOMAIN DATASETS

Dataset	L_{scl} Weight λ_2				
	0.5	1	1.5	2	2.5
IP	81.15	81.58	81.61	**81.72**	81.55
HT	76.24	**76.63**	76.31	76.20	76.07
SA	94.26	**94.34**	94.31	94.30	94.27
LK	96.01	**96.27**	96.18	96.21	96.20

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TABLE X
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%), OA (%), AA (%), AND KC (%) ON IP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	SSRN	DCFSL	HFSL	Gia-CFSL	FSCF-SSL	RPCL-FSL	Ours
1	81.95	85.04	99.01	97.80	**99.76**	93.41	**99.76**	98.54	**99.76**
2	32.38	31.39	42.91	45.14	56.51	47.98	56.59	62.58	**74.24**
3	36.41	30.22	58.84	46.88	53.85	49.35	66.79	57.09	**72.50**
4	56.59	37.82	81.49	75.60	96.38	76.51	**98.06**	93.23	**98.10**
5	67.01	62.50	78.63	67.74	65.15	66.65	71.32	78.16	**80.54**
6	65.39	62.91	95.75	86.50	73.97	84.14	80.70	90.06	**97.05**
7	91.74	92.23	99.55	99.57	99.57	99.57	**100**	**100**	**100**
8	65.35	59.86	76.90	76.03	97.72	83.76	**98.73**	90.51	97.21
9	80.00	95.24	**100**	**100**	**100**	**100**	**100**	**100**	**100**
10	36.71	50.78	60.51	57.08	62.91	58.23	71.39	73.80	**81.21**
11	41.56	51.70	51.14	61.45	70.36	61.32	**78.76**	69.56	73.53
12	27.93	27.27	62.59	48.69	66.85	45.83	68.01	64.40	**82.87**
13	86.55	91.34	99.29	95.35	99.65	96.45	**99.75**	98.50	99.55
14	62.96	59.24	87.98	89.48	89.33	87.83	**91.43**	90.55	88.13
15	29.19	46.15	75.21	67.72	91.57	70.21	**94.83**	80.63	86.75
16	89.43	81.08	97.70	96.82	96.02	96.02	94.55	97.05	**98.18**
OA	46.77 ±2.94	49.02 ±4.18	65.72 ±3.77	64.89 ±3.50	72.22 ±3.22	65.44 ±2.23	77.41 ±1.48	75.11 ±1.97	**81.72 ±1.73**
AA	59.45 ±1.93	60.30 ±2.30	79.22 ±1.61	75.74 ±2.44	82.84 ±2.33	76.08 ±1.48	85.67 ±1.34	84.04 ±1.10	**89.35 ±1.27**
KC	40.55 ±2.95	42.63 ±4.28	61.72 ±3.90	60.36 ±3.96	68.63 ±3.57	61.02 ±2.53	74.40 ±1.64	71.94 ±2.17	**79.36 ±1.93**

datasets with moderate neighborhood class diversity have the best effect when the mask ratio is close to 0.4. The possible reason is that when the neighborhood classes are rich, the probability of containing redundant and noisy pixels in the patch is higher. A high mask ratio enables SCL to better eliminate the interference of redundant and noisy pixels, while when the classes are sparse, the probability of containing redundant and noisy pixels in the patch is lower. A low mask ratio can enable SCL to effectively eliminate the influence of redundant and noisy pixels. In addition, the values of λ_1 are all 2, indicating that modal alignment plays an important role in suppressing model overfitting and improving classification performance. Therefore, it is necessary to emphasize the role of modal alignment more. For the HT, SA, and LK datasets, the optimal value for λ_2 is 1, and for the IP dataset, the optimal value is 2, indicating that SCL based on pixel random masks has a greater effect on IP datasets with rich neighborhood classes.

In summary, by adjusting parameters on η, σ, λ_1, and λ_2 on four publicly available datasets, we can obtain suboptimal solutions on each dataset at a lower cost. This result provides direction and a foundation for further optimizing model performance.

D. Comparison With Other Methods

In order to evaluate the classification performance of the proposed CDFS-CASCL in the case of fewer labeled samples, it is compared with eight related methods, including the traditional machine learning method SVM [2], the conventional deep learning methods 3DCNN [9], SSRN [10] and the cross-domain FSL methods DCFSL [34], HFSL [35], Gia-CFSL [36], FSCF-SSL [22], and RPCL-FSL [37]. The training and testing procedures of these methods are set up as follows. To ensure fair comparison, for the conventional deep learning methods 3DCNN and SSRN, the training and testing classes are the same. Five labeled samples of each class in the target domain dataset are randomly selected as the training set, and the remaining samples are used as the testing set. For the cross-domain FSL methods DCFSL, HFSL, Gia-CFSL, FSCF-SSL, RPCL-FSL, and the proposed CDFS-CASCL method, the target domain is also to randomly select five labeled samples from each class as training samples, and the remaining samples are used for testing. In addition, training on labeled samples from the source domain is also required. For HFSL and FSCF-SSL, the training set of mini-ImageNet is used as the source domain to train the base class. For the DCFSL, Gia-CFSL, RPCL-FSL, and CDFS-CASCL methods, the CH dataset is used as the source domain, and 200 samples in each base class are randomly selected for training. In addition, the HT dataset has fixed training and testing set divisions. We only randomly select five labeled samples of each class from the training set and then test all samples in the testing set. Considering the effect of random sampling, all experiments were performed ten times and then averaged.

Tables X–XIII show the classification results of all methods on the IP, HT, SA, and LK datasets with five labeled samples per class, respectively. Table X shows that the proposed CDFS-CASCL method significantly outperforms all other methods on the IP dataset. Specifically, in the three evaluation dimensions of OA, AA, and KC, the cross-domain FSL methods FSCF-SSL and RPCL-FSL perform outstandingly compared to other comparison methods. The CDFS-CASCL method exceeds the cross-domain FSL method FSCF-SSL by 4.31%, 3.68%, and 4.96%, respectively, and exceeds the RPCL-FSL by 6.61%, 5.31%, and 7.42%, respectively. Furthermore, our method achieved the first place in 11 classes.

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TABLE XI
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%), OA (%), AA (%), AND KC (%) ON HT DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	SSRN	DCFSL	HFSL	Gia-CFSL	FSCF-SSL	RPCL-FSL	Ours
1	81.03	**89.24**	81.94	81.95	70.11	81.24	75.23	81.05	81.29
2	74.54	77.15	**80.72**	77.05	59.67	78.06	66.37	74.09	78.86
3	97.60	95.06	**99.68**	95.01	81.49	92.02	91.33	95.43	99.60
4	86.28	88.15	89.81	89.61	82.25	89.62	86.99	87.27	**92.05**
5	89.06	94.68	95.78	94.88	83.76	94.75	77.55	96.80	**98.31**
6	82.52	72.12	83.57	75.17	86.92	71.12	**87.90**	80.42	82.31
7	55.96	54.53	66.79	57.45	**82.84**	61.75	73.59	77.25	80.71
8	24.85	**42.38**	22.36	41.98	30.73	38.16	23.21	33.99	31.82
9	58.49	59.98	61.50	58.82	56.61	62.03	**76.56**	69.03	71.23
10	47.67	53.29	49.14	47.27	57.16	46.53	**64.45**	47.98	58.40
11	51.04	61.13	**77.97**	53.27	65.09	46.52	65.06	57.07	76.26
12	39.67	44.77	40.70	67.89	56.92	67.16	67.07	68.28	**69.29**
13	19.61	31.56	75.09	79.44	72.60	81.89	69.58	**89.82**	76.39
14	93.48	86.36	**99.80**	84.66	98.38	87.94	96.15	86.52	99.64
15	96.05	96.56	**99.77**	93.00	58.50	91.69	88.69	93.66	99.75
OA	63.75 ±2.26	68.58 ±2.58	70.46 ±1.24	69.95 ±1.74	66.17 ±4.36	69.49 ±2.60	70.29 ±3.36	72.33 ±1.66	**76.63 ±2.03**
AA	66.52 ±1.91	69.80 ±2.61	74.97 ±0.99	73.16 ±1.76	69.53 ±4.92	72.70 ±2.20	73.98 ±3.33	75.91 ±1.52	**79.73 ±1.72**
KC	60.89 ±2.39	65.92 ±2.76	68.08 ±1.33	67.56 ±1.87	63.45 ±4.70	67.05 ±2.80	67.88 ±3.62	70.17 ±1.75	**74.77 ±2.18**

TABLE XII
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%), OA (%), AA (%), AND KC (%) ON SA DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	SSRN	DCFSL	HFSL	Gia-CFSL	FSCF-SSL	RPCL-FSL	Ours
1	96.02	93.08	98.53	99.45	98.45	98.09	**99.92**	98.95	**99.92**
2	94.71	95.99	99.67	98.82	93.58	99.40	95.98	99.09	**99.84**
3	88.56	89.80	96.87	97.31	98.91	95.46	98.73	96.09	**99.05**
4	98.83	98.15	99.31	99.26	98.76	98.52	**99.81**	99.09	99.69
5	94.01	92.34	95.79	91.52	**97.27**	86.55	95.69	90.83	**97.27**
6	98.39	98.80	99.66	**99.84**	99.53	99.25	99.81	98.76	99.82
7	96.61	97.83	**99.93**	99.69	98.12	99.28	99.50	99.17	99.78
8	56.11	59.31	66.17	79.48	80.88	81.31	79.26	82.24	**84.31**
9	94.04	96.93	99.79	99.32	98.87	97.76	99.51	99.94	**99.99**
10	76.44	78.10	90.73	86.78	96.21	83.73	**98.05**	90.00	93.69
11	91.06	89.85	97.34	97.01	99.33	93.56	**99.80**	99.13	98.31
12	99.29	99.34	99.88	99.17	96.39	99.19	97.11	98.36	**99.96**
13	95.51	95.17	**99.01**	97.85	98.41	94.27	98.58	97.09	97.84
14	91.87	91.24	98.13	98.43	96.69	98.46	98.48	97.50	**98.71**
15	64.62	59.06	81.47	75.73	65.01	74.51	78.67	84.59	**88.41**
16	84.08	83.21	94.43	92.75	95.71	90.32	**98.30**	92.62	95.98
OA	81.42 ±2.99	81.74 ±2.31	89.13 ±2.59	90.51 ±0.99	89.71 ±2.31	89.75 ±0.80	91.83 ±1.97	92.33 ±0.78	**94.34 ±0.81**
AA	88.76 ±1.34	88.64 ±1.57	94.79 ±0.55	94.52 ±0.66	94.51 ±1.73	93.10 ±1.19	96.08 ±0.99	95.22 ±0.71	**97.04 ±0.28**
KC	79.44 ±3.23	79.72 ±2.53	87.95 ±2.80	89.45 ±1.09	88.55 ±2.56	88.60 ±0.90	90.92 ±2.19	91.47 ±0.86	**93.71 ±0.89**

Among them, eight classes are fine classes of corn, grass, and soybean, which cover most of the fine classes of the IP dataset. This shows that the CDFS-CASCL method is more advantageous for fine-class classification tasks. It is worth noting that for the classification accuracy of classes 2nd and 12th, the accuracy from the proposed CDFS-CASCL method is more than 10% higher than other optimal methods, which further demonstrates the effectiveness of the proposed method.

Table XI shows the classification results of different classification methods on the HT dataset. Among all comparison methods, the conventional deep learning method SSRN and the cross-domain FSL method RPCL-FSL perform outstandingly compared to other comparison methods. However, the proposed CDFS-CASCL method still outperforms all compared methods in OA. Specifically, in the three evaluation dimensions of OA, AA, and KC, the CDFS-CASCL method exceeded the best-performing conventional deep learning method SSRN by 6.17%, 4.76%, and 6.69%, respectively. It exceeds the best-performing FSL method RPCL-FSL by 4.30%, 3.82%, and 4.60%, respectively, indicating that the

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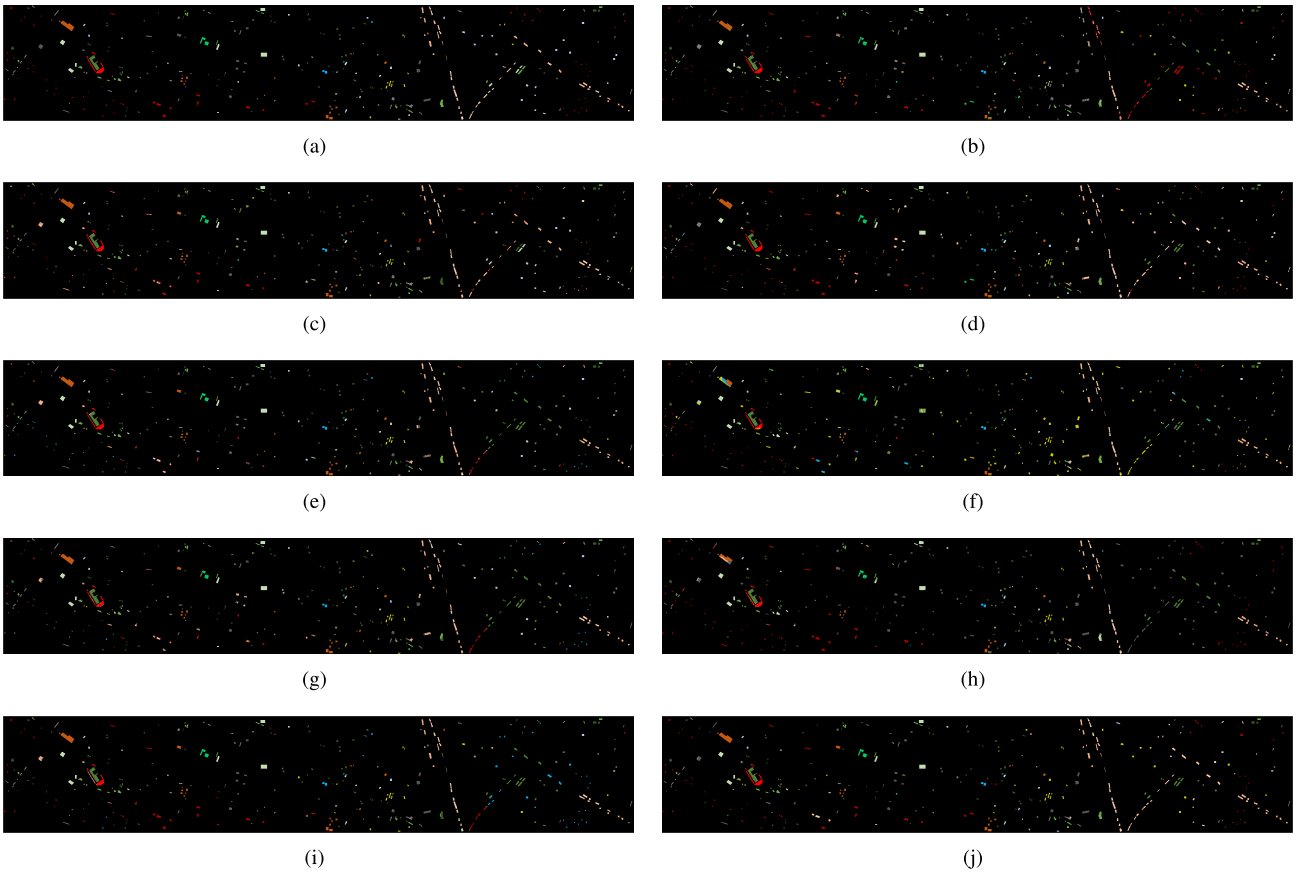


Fig. 11. Classification maps on HT dataset. (a) Ground-truth, (b) SVM, (c) 3DCNN, (d) SSRN, (e) DCFSL, (f) HFSL, (g) Gia-CFSL, (h) FSCF-SSL, (i) RPCL-FSL, and (j) Ours. Please enlarge the images to see the difference.

TABLE XIII
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%), OA (%), AA (%), AND KC (%) ON LK DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	SSRN	DCFSL	HFSL	Gia-CFSL	FSCF-SSL	RPCL-FSL	Ours
1	77.18	85.93	97.69	98.69	96.63	98.41	98.04	97.60	**99.27**
2	53.19	35.45	78.07	88.93	**96.56**	89.41	95.28	88.89	91.60
3	72.73	78.83	89.19	92.02	86.16	91.18	90.47	91.06	**92.53**
4	63.97	65.35	85.53	92.26	85.84	90.94	89.05	93.69	**94.13**
5	66.36	72.68	92.89	93.83	96.02	93.60	**98.17**	86.85	96.50
6	84.16	84.54	94.18	93.22	91.12	91.36	92.86	89.38	**95.28**
7	99.96	99.94	99.86	99.85	99.92	99.87	**99.97**	99.09	99.39
8	80.08	73.27	84.63	81.08	88.13	81.71	**90.57**	81.20	84.64
9	43.22	41.74	82.10	80.06	**92.82**	77.71	91.05	87.17	90.02
OA	78.93 ±3.63	80.07 ±2.50	92.56 ±2.53	95.08 ±1.18	93.49 ±2.45	94.49 ±1.72	94.93 ±1.96	94.89 ±2.22	**96.27 ±2.12**
AA	71.20 ±3.09	70.86 ±2.43	89.35 ±2.96	91.10 ±2.18	92.58 ±1.66	90.46 ±2.32	**93.94 ±1.74**	90.55 ±2.82	93.71 ±2.44
KC	73.41 ±4.27	74.74 ±3.07	90.39 ±3.21	93.58 ±1.52	91.59 ±3.09	92.82 ±2.21	93.43 ±2.49	93.35 ±2.84	**95.15 ±2.72**

prototype learned by CDFS-CASCL is more accurate than RPCL-FSL.

Table XII shows the classification results of different classification methods on the SA dataset, and the proposed CDFS-CASCL method still outperforms all compared methods. Specifically, in the three evaluation dimensions of OA, AA, and KC, the CDFS-CASCL method exceeds the best-performing cross-domain FSL method RPCL-FSL by 2.01%, 1.82%, and 2.24%, respectively, and exceeds the best-performing conventional deep learning methods SSRN by 5.21%, 2.25%, and 5.76%, respectively. Moreover, the CDFS-CASCL method achieved the best classification accuracy in both the 8th and 15th classes, the most difficult ones to classify, which further confirms the superiority of the proposed method. In addition, we can see that the classification indicators of the improved prototype-based cross-domain FSL

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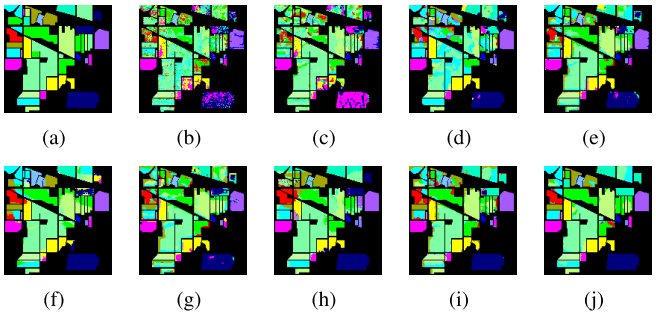


Fig. 12. Classification maps on IP dataset. (a) Ground-truth, (b) SVM, (c) 3DCNN, (d) SSRN, (e) DCFSL, (f) HFSL, (g) Gia-CFSL, (h) FSCF-SSL, (i) RPCL-FSL, and (j) Ours. Please enlarge the images to see the difference.

methods RPCL-FSL and CDFS-CASCL are significantly better than other comparison methods. This indicates that learning an accurate image prototype representation with strong generalization ability through a small number of labeled samples is crucial to the final classification result.

Table XIII shows the classification results of different classification methods on the LK dataset, and the proposed CDFS-CASCL method still outperforms all compared methods. Specifically, in the three evaluation dimensions of OA, AA, and KC, the CDFS-CASCL method exceeds the best-performing cross-domain FSL method DCFSL by 1.19%, 2.61%, and 1.57%, respectively, and exceeds the best-performing conventional deep learning method SSRN by 3.71%, 4.36%, and 4.76%. Moreover, the CDFS-CASCL method achieved the best classification accuracy among the four classes, which illustrates the superiority of the proposed method. In addition, we can see that on the UAV-borne dataset LK, the classification indicators of the cross-domain few-shot method are generally better than other comparison methods. This shows that when the domain differences are large, transferring knowledge from the source domain is beneficial to achieving better classification performance in the target domain.

In order to compare the classification performance of each comparison method more intuitively when there are five labeled samples of each class in the target domain, we show the classification maps of each comparison method on the IP, HT, SA, and LK datasets, as well as the corresponding ground-truth. As shown in Figs. 11–14, we can see that the classification map of the proposed CDFS-CASCL method is the closest to Ground-truth and has the least misclassification between classes. This shows that our model has a more accurate understanding of class concepts and therefore generates a more discriminative class feature distribution in the feature space.

To evaluate the classification performance of the proposed CDFS-CASCL method under different numbers of labeled samples per class, we randomly select 1 to 5 labeled samples from each class in the target domain for model training, and the remaining samples are for testing. The classification results are still based on the average of ten experimental results. Fig. 15 shows the OA of all methods on the IP, HT, SA, and LK datasets under different numbers of labeled samples in each class. First, we can see that as the number of labeled samples

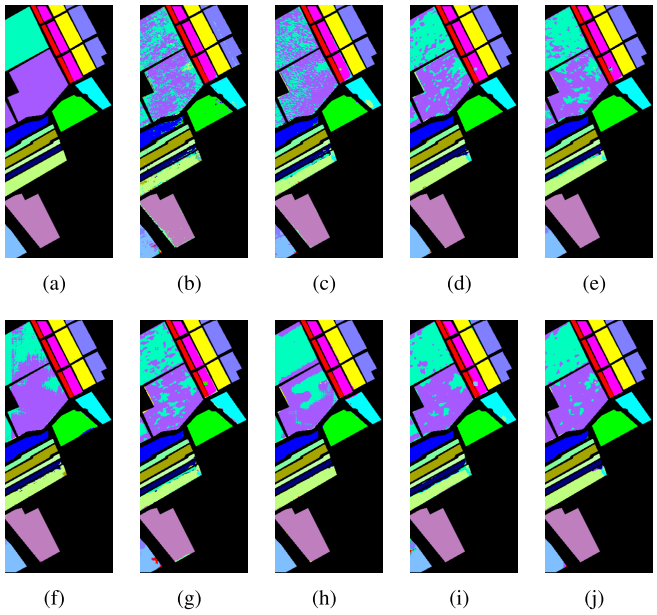


Fig. 13. Classification maps on SA dataset. (a) Ground-truth, (b) SVM, (c) 3DCNN, (d) SSRN, (e) DCFSL, (f) HFSL, (g) Gia-CFSL, (h) FSCF-SSL, (i) RPCL-FSL, and (j) Ours. Please enlarge the images to see the difference.

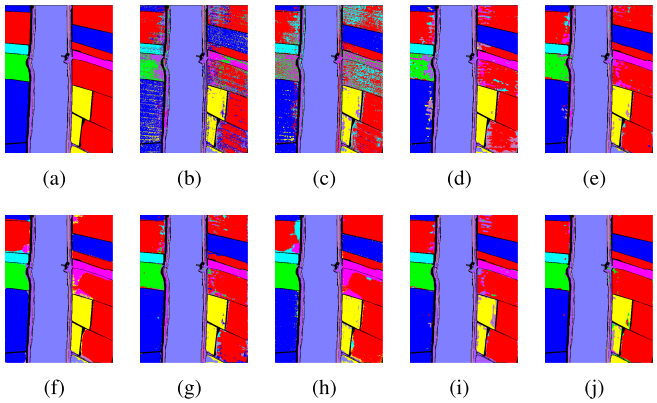


Fig. 14. Classification maps on LK dataset. (a) Ground-truth, (b) SVM, (c) 3DCNN, (d) SSRN, (e) DCFSL, (f) HFSL, (g) Gia-CFSL, (h) FSCF-SSL, (i) RPCL-FSL, and (j) Ours. Please enlarge the images to see the difference.

increases, the OA of all methods shows an upward trend. It is worth noting that the CDFS-CASCL achieved the best results in different datasets even when with different labeled samples. Second, we find that the performance leadership of the IP dataset is more prominent than other datasets. Second, we found that the performance leadership of IP and HT datasets is more prominent than other datasets. Comparing the best-performing methods on the IP and HT datasets, our method yielded an average OA improvement of more than 5% and 4% with different numbers of labeled samples in each class. This shows that our model still learns more accurate class prototypes with stronger generalization ability when the number of samples is extremely scarce. In addition, the stable high-performance performance on these four datasets also illustrates the robustness of our method to different numbers of labeled samples.

In addition, in order to visually demonstrate the feature extraction performance of different methods, we used t-SNE technology to demonstrate the 2-D projection features of

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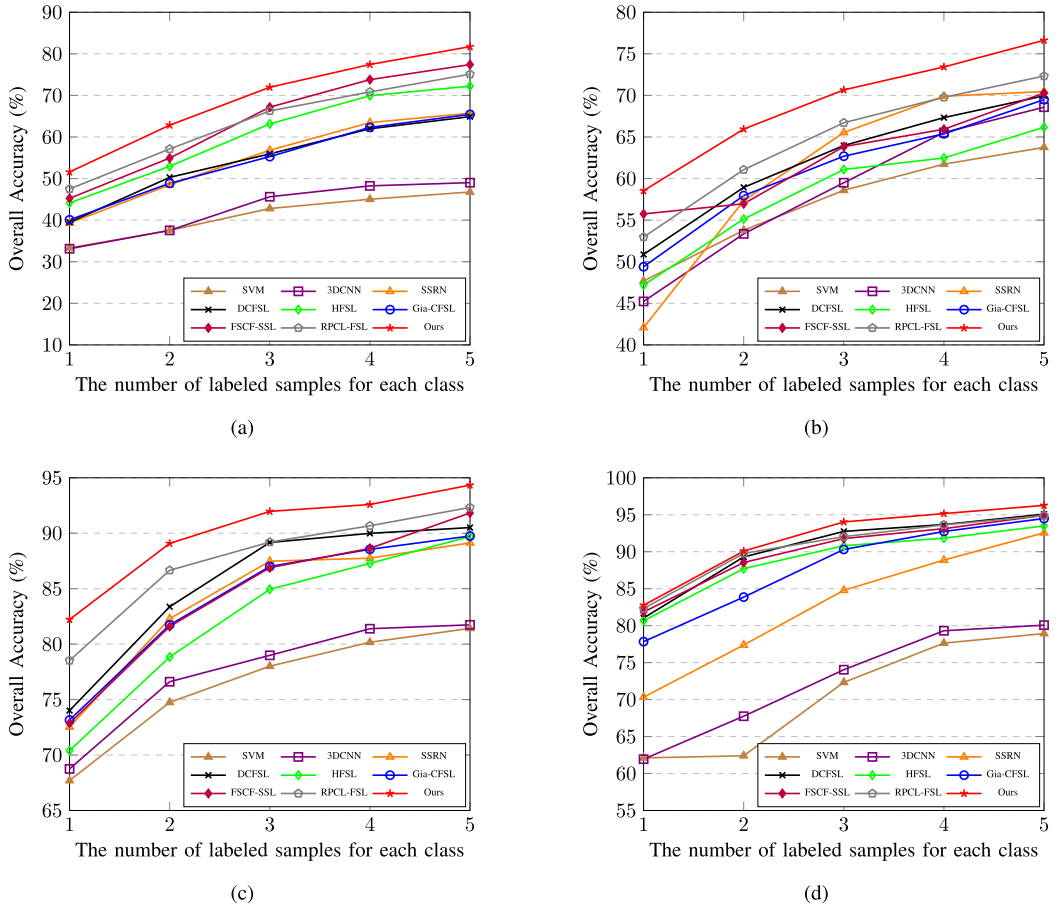


Fig. 15. OA (%) of all methods under different numbers of labeled samples per class in the target domain dataset: (a) IP, (b) HT, (c) SA, and (d) LK.

three cross-domain FSL methods (Gia-CFSL, RPCL-FSL, and CDFS-CASCL) on the IP, HT, SA, and LK datasets in Fig. 16. Obviously, compared with Gia-CFSL and RPCL-FSL methods, the proposed method CDFS-CASCL has the least confusion of different class features on the four datasets, demonstrating the best classification boundary, indicating that the learned features are more separable. For example, on the IP dataset, it is difficult to distinguish the 10th class "Soybean note," the 11th class "Soybean mint," and the 12th class "Soybean mint" due to their similar land cover classes. From Fig. 16(a)–(d), it can be seen that the proposed method can effectively distinguish these three classes. Similarly, on the SA dataset, the 8th class "Grapes untrained" and the 15th class "Vinyard untrained" are the most difficult to distinguish classes. From Fig. 16(g)–(i), it can be seen that our method has the least feature confusion and exhibits the best feature separability. On the LK dataset, it can be seen from Fig. 16(j)–(l) that compared to other methods, the 6th, 8th, and 9th classes of our method are more compact, and there is almost no feature confusion. In addition, on the HT dataset, it can be seen from Fig. 16(d)–(f) that the feature distribution of classes is relatively loose. In this case, our method effectively preserves intraclass differences while minimizing interclass confusion. In summary, our proposed CDFS-CASCL method effectively improves the separability of features in the feature space.

TABLE XIV
ABLATION COMPARISON RESULTS (%) OF EACH VARIANT OF CDFS-CASCL

Module			Dataset			
FS	CA	SCL	IP	HT	SA	LK
✓			75.43	74.34	93.11	95.49
✓	✓		79.72	75.67	93.93	95.98
✓		✓	81.01	75.72	93.94	96.04
✓	✓	✓	**81.72**	**76.63**	**94.34**	**96.27**

E. Ablation Study

The main contribution of the proposed CDFS-CASCL lies in the CA and SCL methods. To better analyze the role of these two methods, we evaluated their contributions through ablation experiments. As shown in Table XIV, we focus on investigating four situations: 1) only doing FSL on the source domain and target domain; 2) adding CA on the basis of 1); 3) adding SCL on the basis of 1); and 4) add CA and SCL on the basis of 1).

From Table XIV, we can see that 2) adding CA on the basis of 1) increases the OA on the IP, HT, SA, and LK datasets by 4.29%, 1.33%, 0.82%, and 0.49%, respectively. This shows that CA allows image features to obtain cross-modal information. 3) adding SCL on the basis of 1) improves the IP, HT, SA, and LK datasets by 5.58%, 1.38%, 0.83%,

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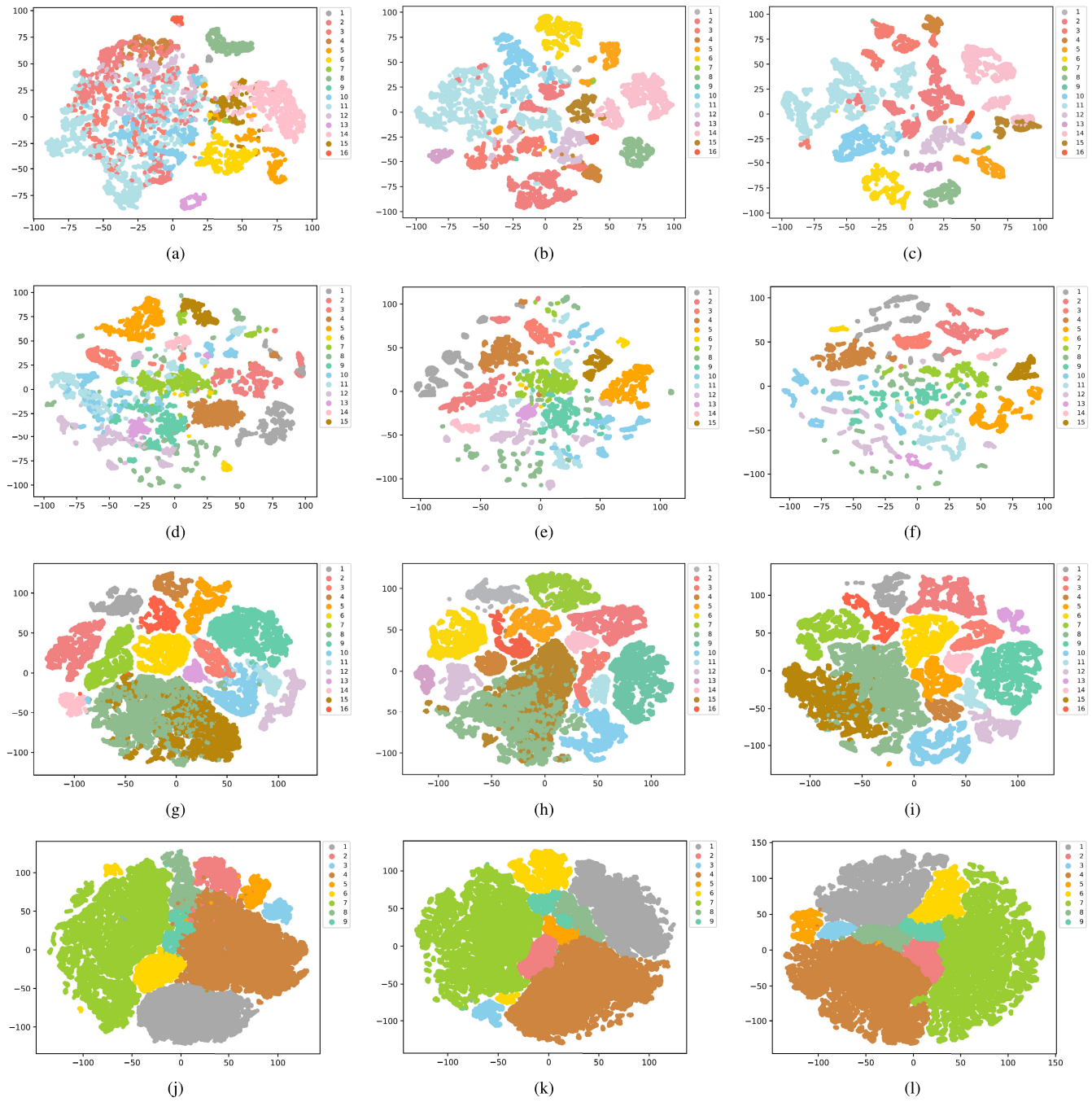


Fig. 16. Two-dimensional feature visualization on four datasets. (a) Gia-CFSL features on IP, (b) RPCL-FSL features on IP, (c) CDFS-CAFSL features on IP, (d) Gia-CFSL features on HT, (e) RPCL-FSL features on HT, (f) CDFS-CAFSL features on HT, (g) Gia-CFSL features on SA, (h) RPCL-FSL features on SA, (i) CDFS-CAFSL features on SA, (j) Gia-CFSL features on LK, (k) RPCL-FSL features on LK, and (l) CDFS-CAFSL features on LK.

and 0.55%, respectively. This shows that SCL can improve the distinction between different classes in the target domain in the metric space and reduce the model's overfitting of training samples. According to 4), it can be seen that CA and SCL work together to further improve the model classification performance based on 2) and 3).

F. Computational Complexity Analysis

As shown in Table XV, we evaluate the computational complexity of the model based on its training time, testing time, the number of floating-point operations (FLOPs), and the number of parameters. However, for the traditional machine learning method (SVM) and the conventional deep learning methods (3DCNN and SSRN), the training time only includes the time for training a single domain with limited labeled samples. For most cross-domain FSL methods (DCFSL, Gia-CFSL, RPCL-FSL, and CDFS-CASCL), the training time includes source domain training time and target domain fine-tuning time. Specifically, for HFSL and FSCF-SSL, in addition to the time required for cross-domain FSL methods mentioned above, it also includes model transmission time. From Table XV, it can be seen that conventional

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TABLE XV
COMPUTATION TIME (S), FLOPS (M), AND PARAMETERS (M) ON EACH DATASET WITH ALL METHODS

Dataset	Metric	Method								
		SVM	3DCNN	SSRN	DCFSL	HFSL	Gia-CFSL	FSCF-SSL	RPCL-FSL	Ours
IP	Training time	1.66	90.82	122.92	1939.09	731.28	3308.59	1129.44	435.17	1075.49
	Testing time	0.10	3.91	8.78	3.28	5.22	1.46	4.78	1.44	1.07
	FLOPs	-	0.20	1.38	47.28	400.45	130.82	400.45	47.37	29.07
	#Params	-	0.10	0.35	4.28	6.27	0.86	6.28	4.27	0.31
HT	Training time	1.76	87.11	74.77	1382.88	608.13	2528.23	1017.78	333.24	868.67
	Testing time	0.09	3.94	7.26	3.13	4.38	1.51	4.31	1.55	1.13
	FLOPs	-	0.14	1.04	46.83	399.94	124.86	399.94	46.92	28.80
	#Params	-	0.07	0.26	4.27	6.19	0.85	6.19	4.26	0.30
SA	Training time	1.89	52.24	181.68	2053.16	792.65	3614.00	2436.57	581.66	1130.75
	Testing time	0.53	10.64	43.29	12.64	25.97	8.36	25.52	8.94	6.24
	FLOPs	-	0.21	1.41	47.31	400.48	130.85	400.48	47.40	29.09
	#Params	-	0.10	0.35	4.28	6.28	0.86	6.28	4.27	0.31
LK	Training time	1.77	53.20	594.70	1685.89	1184.31	4294.49	2167.80	1032.62	1002.92
	Testing time	1.28	47.75	250.88	63.82	117.36	66.96	116.12	50.06	32.79
	FLOPs	-	0.27	1.81	47.85	401.08	93.12	401.08	47.94	29.42
	#Params	-	0.14	0.45	4.28	6.38	0.86	6.38	4.28	0.32

deep learning methods require much less training time and FLOPs than cross-domain FSL methods, indicating that FSL methods based on cross-domain knowledge transfer pay higher computational costs while achieving better classification performance. Meanwhile, it can be observed that HFSL, RPCL-FSL, and CDFS-CASCL spend less time compared to other cross-domain FSL methods. Furthermore, our proposed CDFS-CASCL achieves better classification performance with fewer FLOPs and parameters compared to HFSL and RPCL-FSL at a similar time cost.

IV. CONCLUSION

This article proposes the CDFS-CASCL method based on CA and SCL for few-shot HSI classification. A CA strategy is proposed to address the limited prototype representation capabilities of image modalities. The semantic features of labels are extracted through a pretrained language model, which is used to guide the learning of image prototype representation and improve the semantic representation capabilities of the image prototypes. In addition, the proposed method also integrates an SCL strategy and uses a neighborhood pixel mask as an augmentation method to enable the model to learn the mask-robust discriminative features and suppress the adverse effects of redundant and noisy pixels. The two strategies work together to improve the accuracy of model feature expression, thereby significantly improving the model's discriminability in the case of sample scarcity. Extensive experimental results based on IP, HT, SA, and LK datasets show that the proposed method significantly outperforms existing methods.

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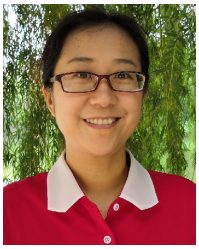
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